Hamza Mehmood

.NET Software Engineer

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PROFESSIONAL SUMMARY

.NET software engineer with 4+ years of experience building production backend systems in C#, ASP.NET Core and Azure: event-driven microservices, high-performance SQL, and secure, well-tested REST APIs. Cut enterprise report generation from 180 seconds to 32 seconds, reduced production defects by 91% across a three-client platform, and shipped a production LLM chatbot at under 3-second latency with 30% lower API cost.

TECHNICAL SKILLS

Languages and CS: C#, VB.NET, SQL, Python (Django, Flask), TypeScript, JavaScript, Java, C++, OOP, Data Structures

Backend and APIs: .NET 8, .NET Core, ASP.NET Core, ASP.NET MVC, Web API, Entity Framework Core, Dapper, ADO.NET, REST APIs, SOAP/WCF, Microservices, Monolith Migration, Event-Driven Architecture, OAuth 2.0, JWT, Payment Processing

Cloud and DevOps: Microsoft Azure (Functions (Durable), Logic Apps, API Management, Service Bus, App Service, Blob Storage, Key Vault, Entra ID), Docker, Terraform, Azure DevOps, GitHub Actions, CI/CD, Datadog, Azure Application Insights

Databases: SQL Server (SSMS), Azure SQL, PostgreSQL, Oracle (PL/SQL, ODBC), Cosmos DB, Stored Procedures, Data Modeling, Schema Design, Indexing, Query Optimization, Execution Plan Analysis, Power BI

Testing and Practices: xUnit, MSTest, Postman, Unit and Integration Testing, UML, Clean Architecture, Code Review, SDLC, SOLID, Design Patterns, Clean Code, Agile/Scrum, Git (CLI), GitHub, Visual Studio, Technical Documentation

Frontend: WPF (MVVM), Telerik UI for WPF, Angular, TypeScript, HTML5, CSS3, Blazor

AI and LLM: LLM APIs (OpenAI, Azure OpenAI), Azure AI Foundry, LangChain, OpenRouter, MCP, RAG, Vector Databases, Prompt Engineering, LLM-as-a-Judge Evaluation, AI-Assisted Dev (Claude Code, Codex, Copilot, ChatGPT, Perplexity)

PROFESSIONAL EXPERIENCE

Research Assistant

Texas Tech University, Department of Computer Science, Lubbock, TX | Aug 2026 - Present

- Research automated feedback generation, comparing single-prompt and multi-agent LLM orchestration across programming, essay and mathematics domains.
- Build LLM-as-a-judge evaluation frameworks that score generated output against rubric criteria for precision, correctness and consistency.
- Run controlled Python experiments against open-source models (DeepSeek R1, Qwen) with reproducible run configuration and result capture.

Graduate Assistant, Business Technology Support

Texas Tech University - TTU Online, Lubbock, TX | Feb 2026 - Aug 2026

- Shipped a production LLM chatbot backed by a vector database, cutting response latency to under 3 seconds and API costs by 30% after a full-context design proved unscalable.
- Built agents for live workflows: a Flask-based Twilio messaging agent (sends, replies, opt-out suppression, delivery tracking) and a content-change monitoring agent. Built an AI token-consumption tracker.
- Engineered an evaluation pipeline pairing a third-party API with a Python NLP analyzer, improving output-accuracy validation by 20% against real conversation logs.
- Built a Python data pipeline with mapping and validation logic for record migrations across 40 school districts, cutting manual effort by 60%.

Consultant .NET

Systems Limited (PartnerLinQ), Lahore, Pakistan | May 2024 - Jul 2025

- Served as primary production responder for a platform integrating four downstream systems, resolving 30-40 incidents per month at 70% SLA compliance and reducing recurring failures by 30%.

- Designed event-driven services on Azure Functions, Logic Apps, API Management and Service Bus with retry, idempotency and dead-letter handling.
- Managed production environment configuration with Terraform; packaged and deployed services with Docker through Azure DevOps pipelines.
- Set up Datadog monitors and alerts and used Azure Application Insights to track cross-service failures; traced unresponsive apps, missing records and reconciliation mismatches to root cause.
- Implemented OAuth 2.0 and Azure Key Vault secrets management; authored about 9 XML/XSD schema mappings per month between partner systems.

Software Engineer
NETSOL Technologies, Lahore, Pakistan | Jun 2022 - Apr 2024

- Delivered 10+ modules on an enterprise lease and finance platform serving three clients, owning requirements through production release.
- Cut a reporting module from 180 seconds to 32 seconds by profiling execution plans, eliminating table scans across two large joins, and redesigning stored procedures with indexed views.
- Led code reviews and mentored junior developers; production defects fell 91% and sprint delivery held at 94%.
- Worked in ASP.NET MVC and WPF with Telerik controls: built a WPF/MVVM banking channel module over WCF services, and Angular data grids and workflow forms used daily by client operations staff.

.NET Core Developer
Hawk Logix, Lahore, Pakistan | Oct 2021 - Apr 2022

- Built 80+ REST endpoints in Clean Architecture for a US medical billing platform (patient records, claims, billing, insurance workflows) processing about 30K records per month.

PROJECTS

- **OrderFlow, Event-Driven Order Processing** (C#, ASP.NET Core, EF Core) | github.com/hamzamehmood46/orderflow-dotnet: idempotent API, exponential-backoff retries and dead-letter handling; 8 concurrent duplicate requests create exactly one order. Live demo
- **Transactional Outbox and Idempotent Inbox** (C#, EF Core) | github.com/hamzamehmood46/outbox-pattern-dotnet: atomic event publishing with competing dispatchers, crash recovery and exactly-once handler execution.
- **Claims API, Medical Billing** (C#, ASP.NET Core, JWT, Clean Architecture) | github.com/hamzamehmood46/claims-api-dotnet: role-based claim workflow with a domain-enforced state machine. Live demo
- **Multi-Tenant SaaS API** (C#, ASP.NET Core, EF Core) | github.com/hamzamehmood46/tenant-saas-api-dotnet: tenant isolation in query filters and a write guard, roles, per-tenant rate limits; 28 tests. Live demo
- **Employee Directory** (Angular, ASP.NET Core, EF Core) | github.com/hamzamehmood46/employee-directory-angular-dotnet: live async email validation, server-side paging; 65 tests in CI.
- **Event-Sourced Ledger with CQRS** (C#, ADO.NET) | github.com/hamzamehmood46/event-sourced-ledger-dotnet: optimistic concurrency; parallel withdrawals can never overdraw an account.
- **AI and LLM Projects** (Python) | github.com/hamzamehmood46/WriteDetect: LLM-as-a-judge lecture scoring, AI-vs-human text detector (live demo).

EDUCATION

Master of Science, Computer & Information Sciences
Texas Tech University, Lubbock, TX | Aug 2026 | GPA 3.3/4.0

Bachelor of Science, Software Engineering
National Textile University, Pakistan | 2021 | GPA 3.5/4.0

CERTIFICATIONS

IBM Back-End Developer Professional Certificate, Coursera, 2023